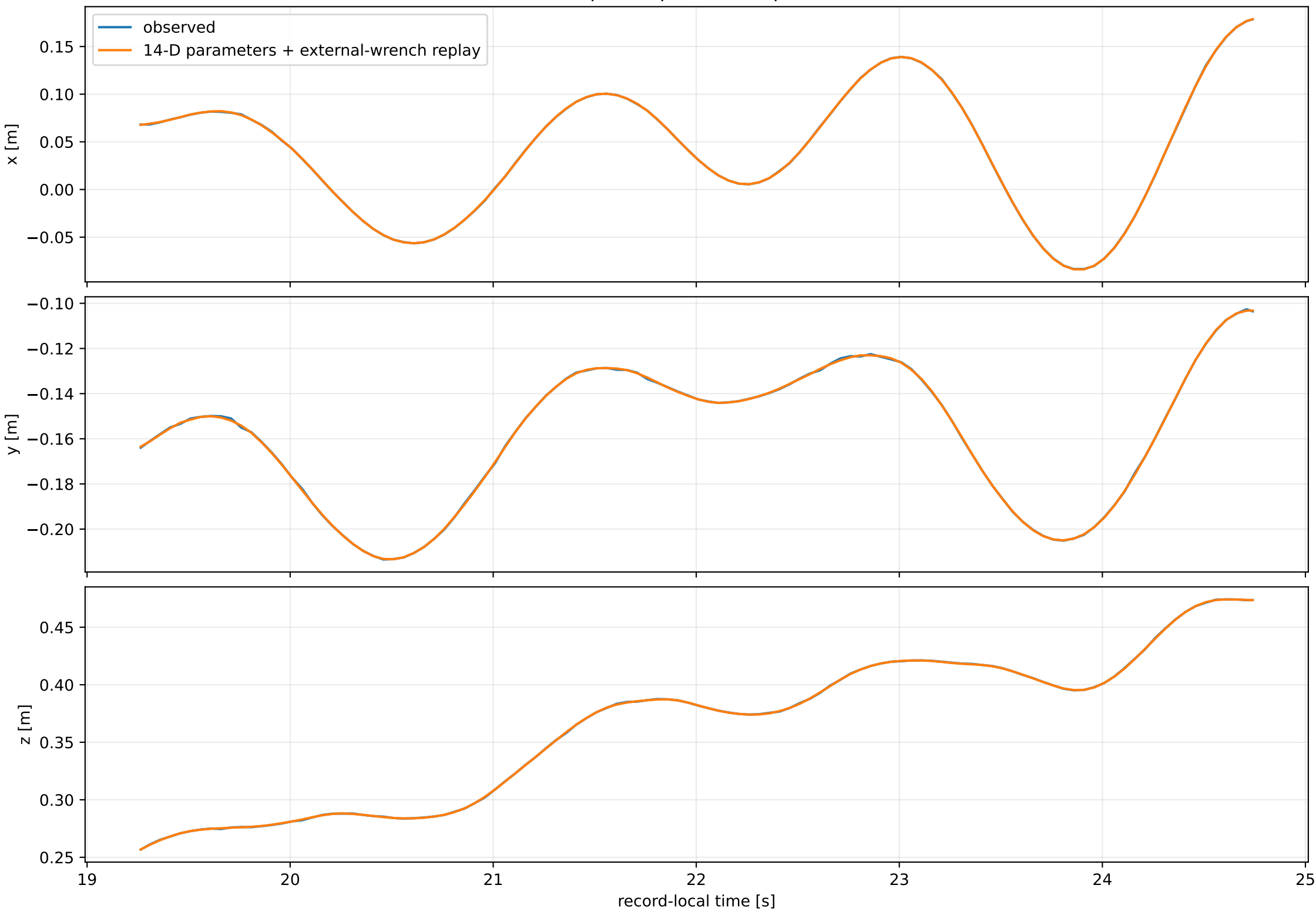
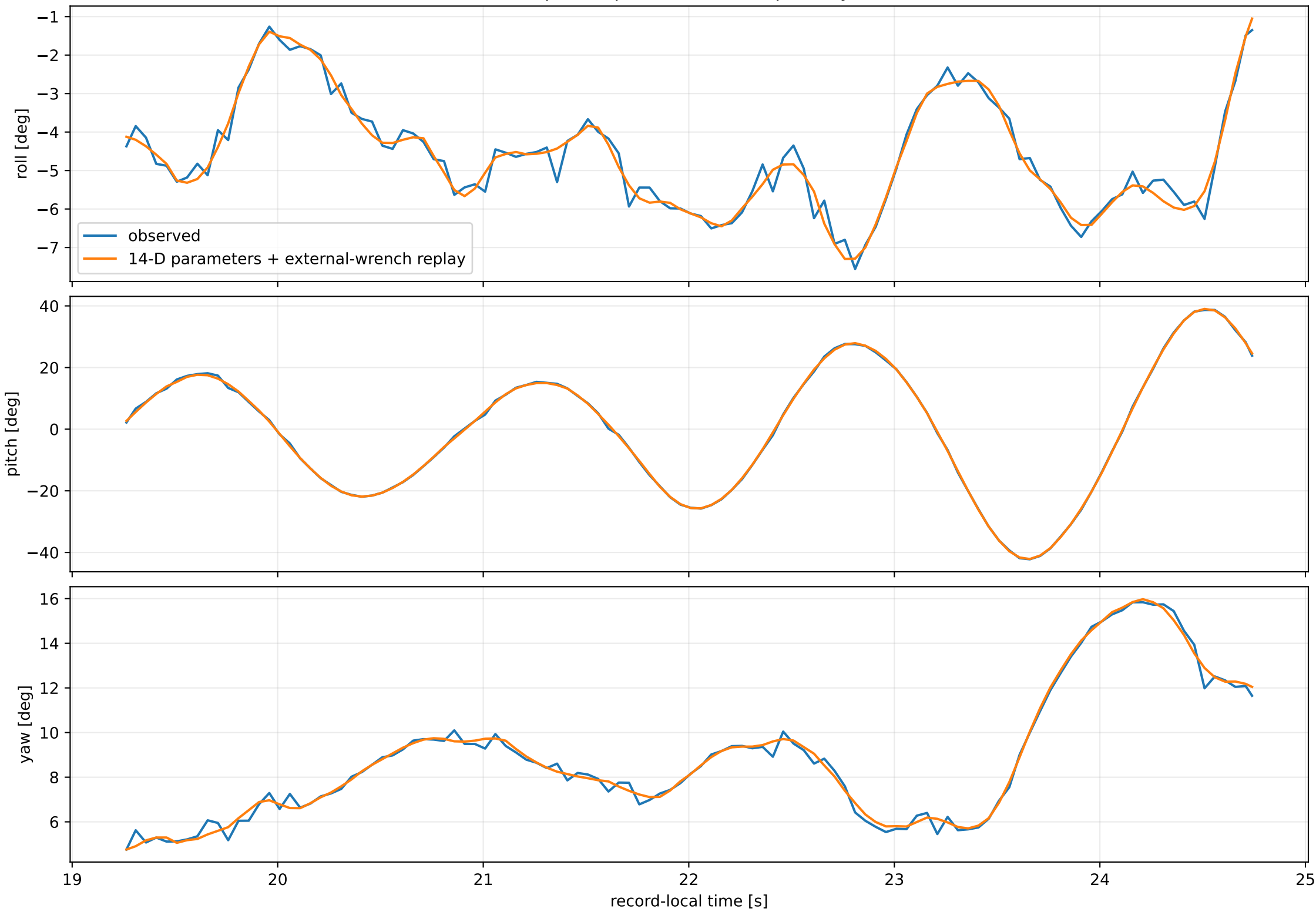


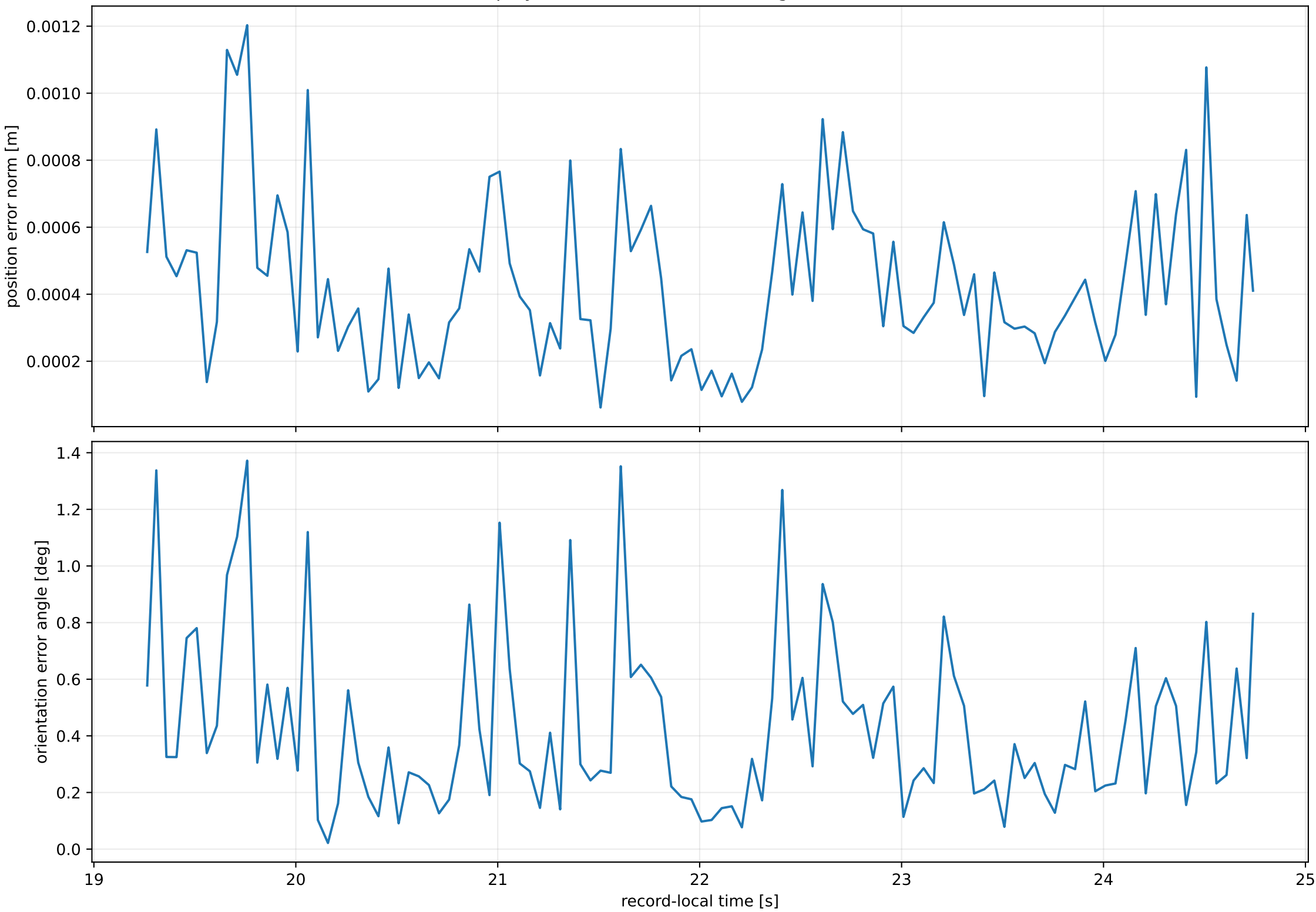
Fine pose reproduction: position



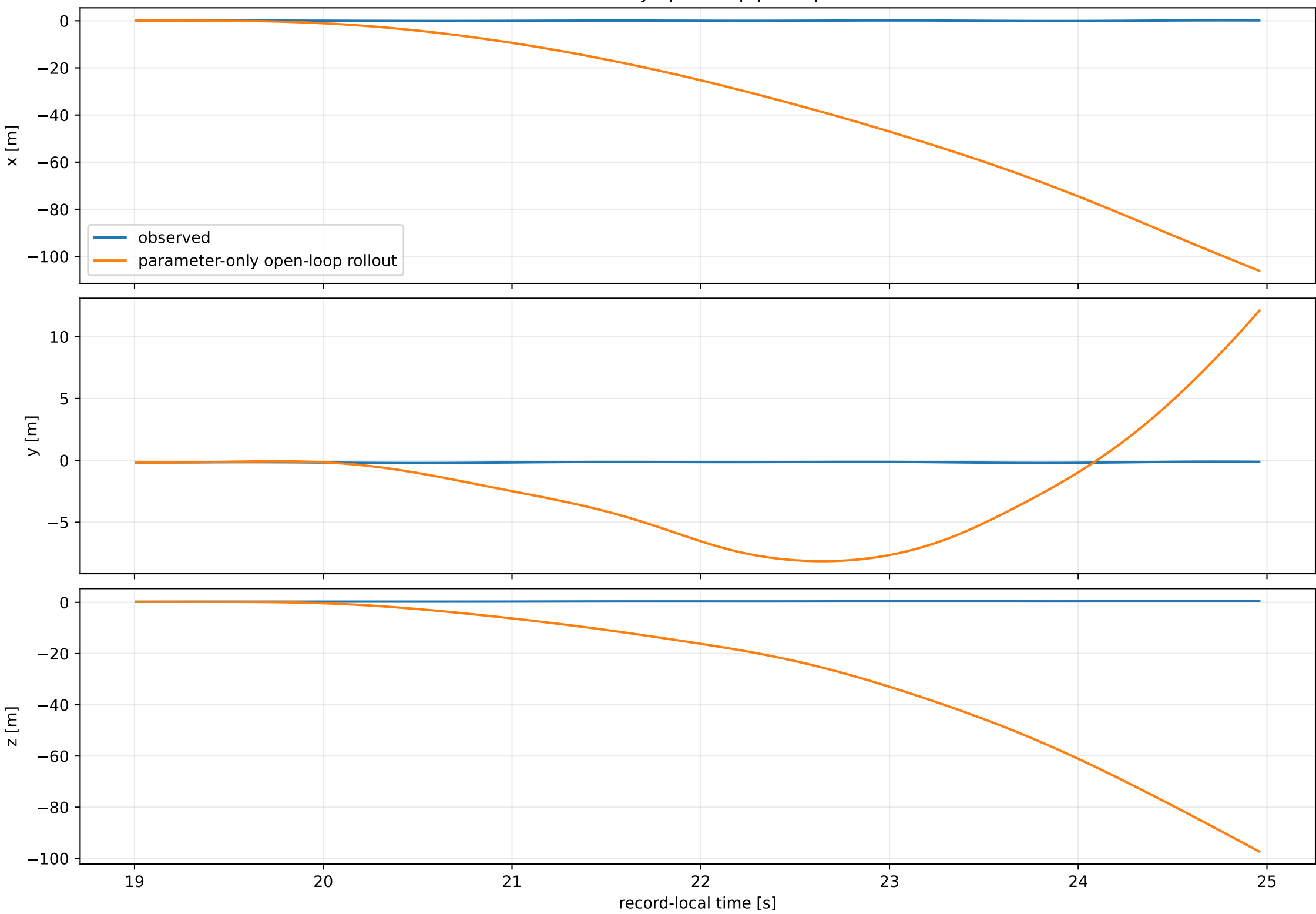
Fine pose reproduction: roll / pitch / yaw



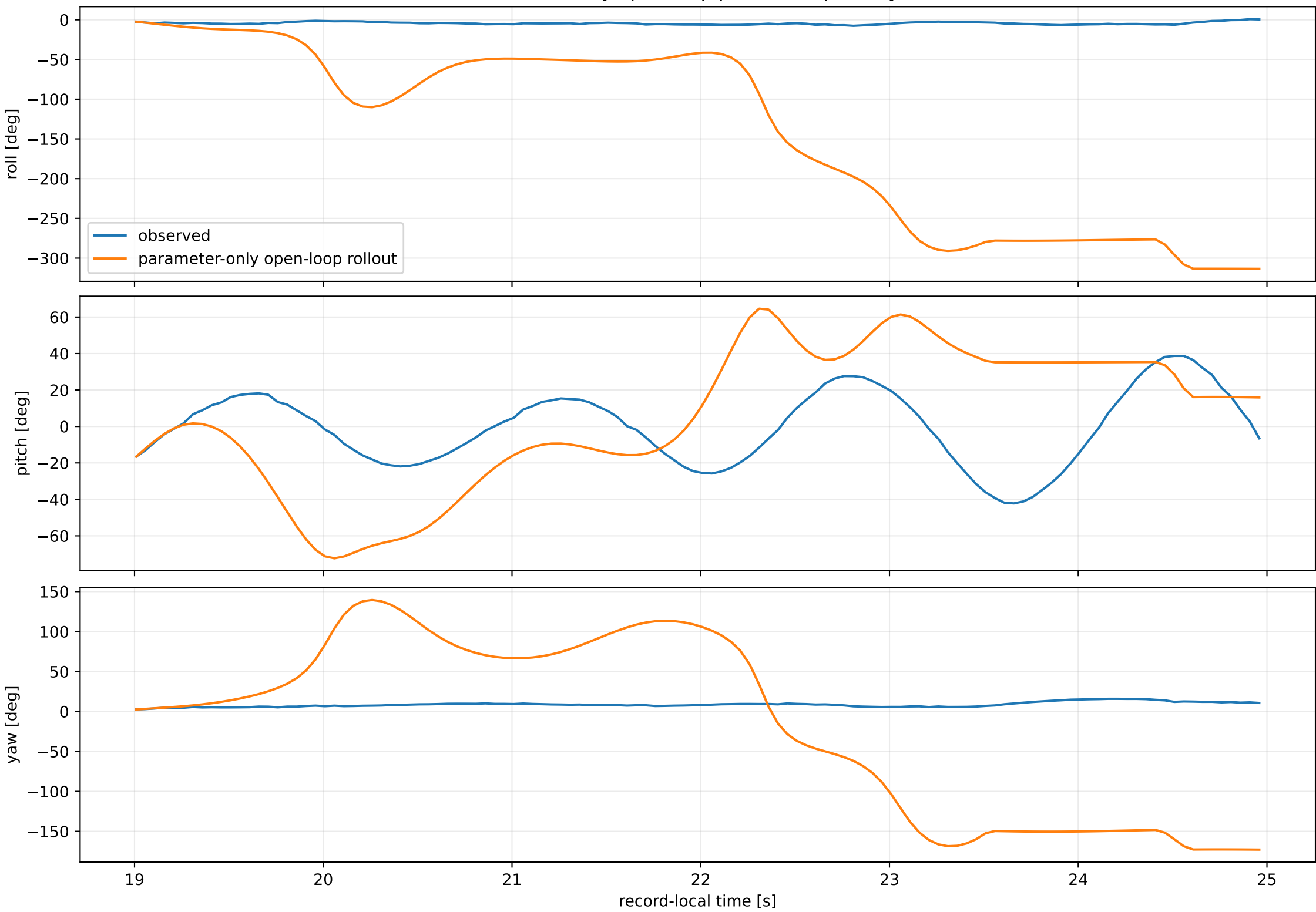
Fine replay error after the 14-D change — lower is better



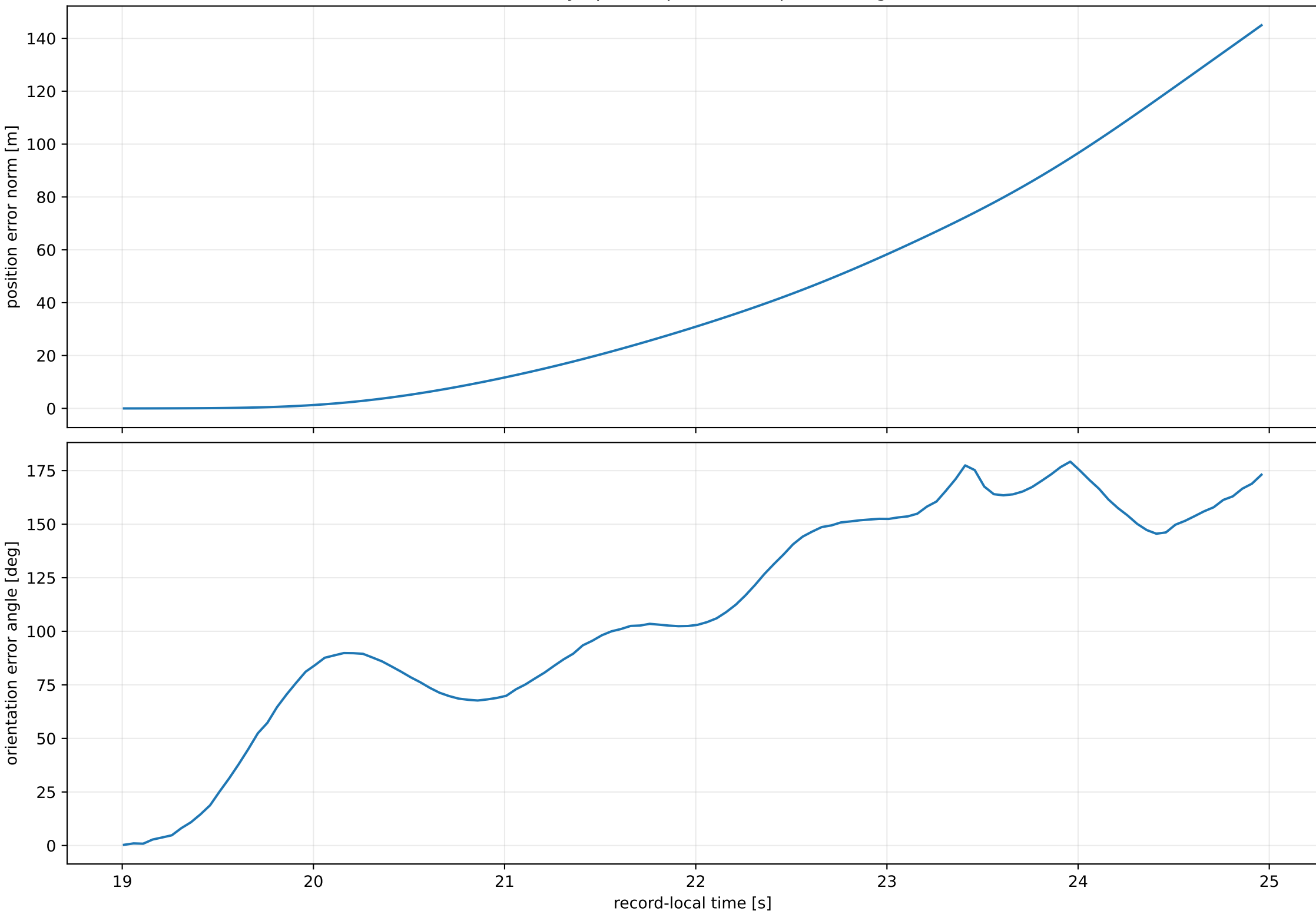
Parameter-only open-loop pose: position



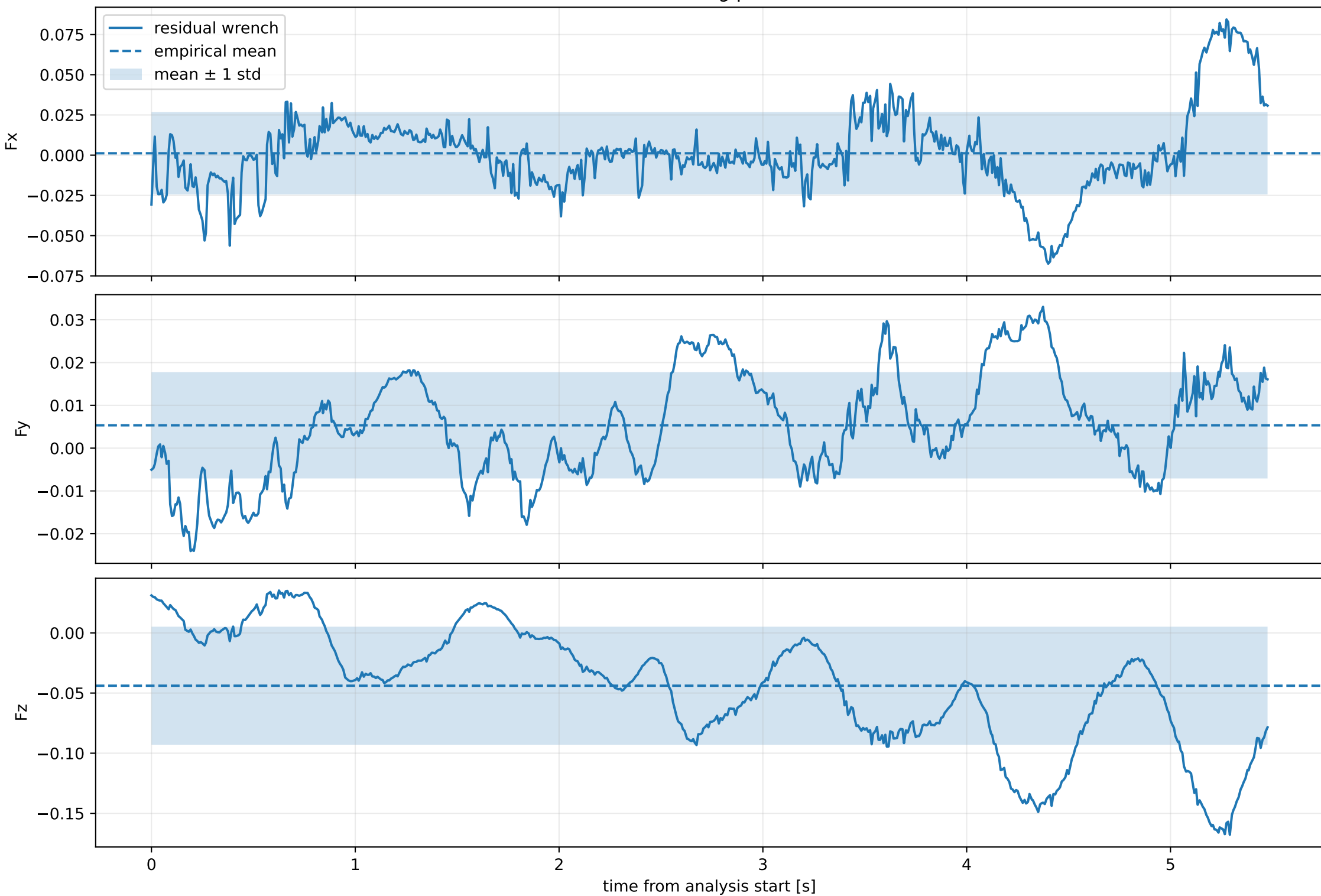
Parameter-only open-loop pose: roll / pitch / yaw



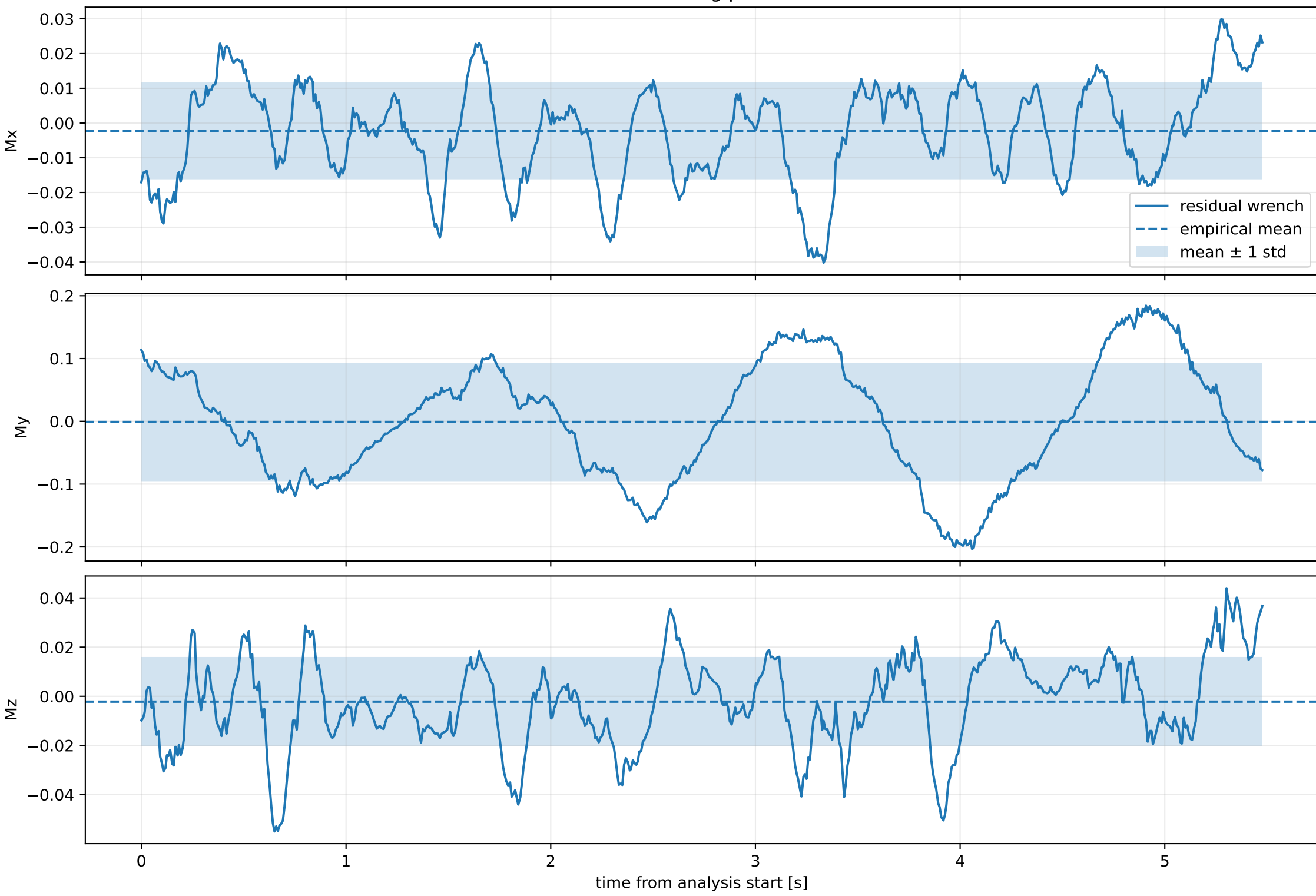
Parameter-only open-loop rollout (separate diagnostic)



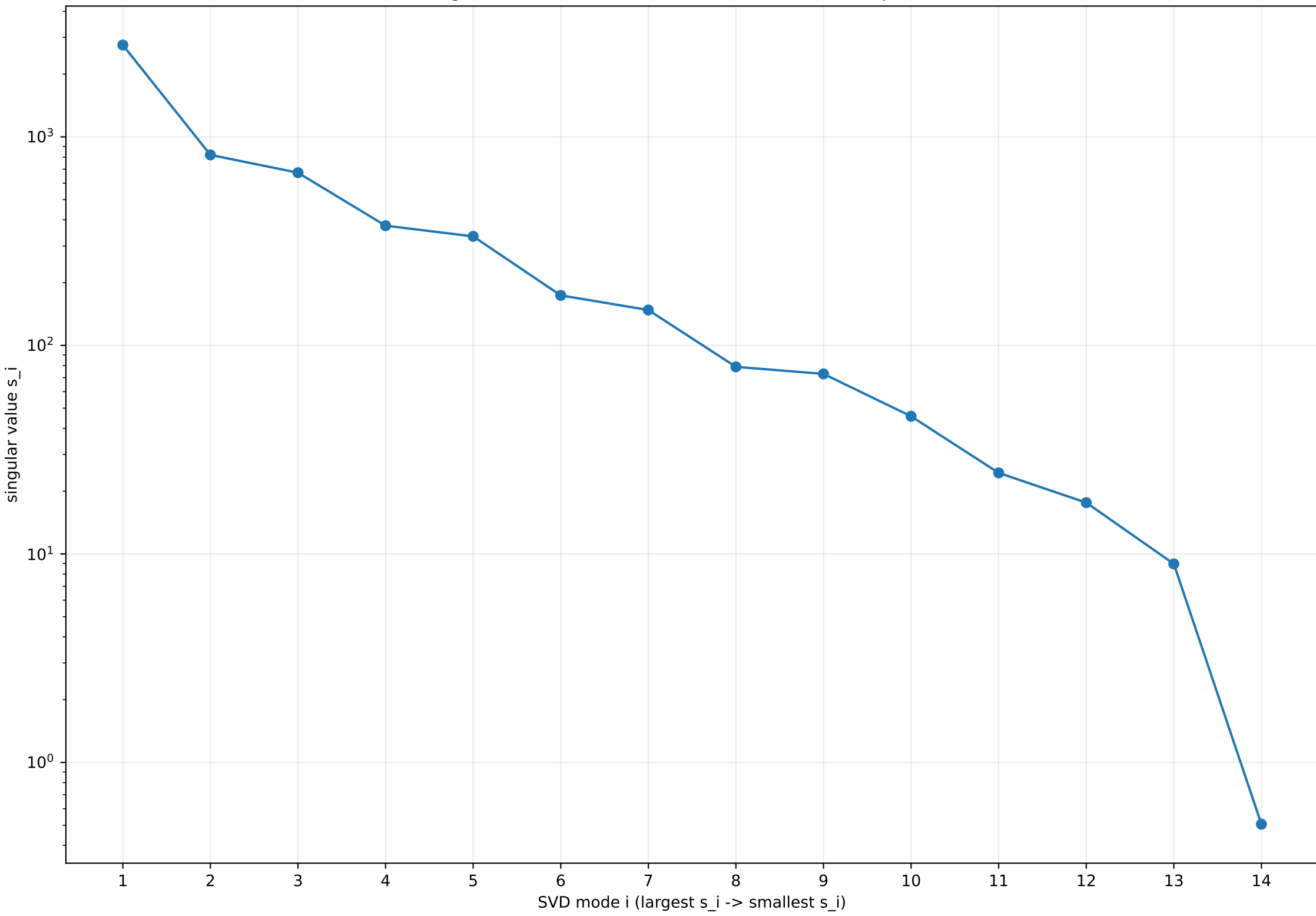
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



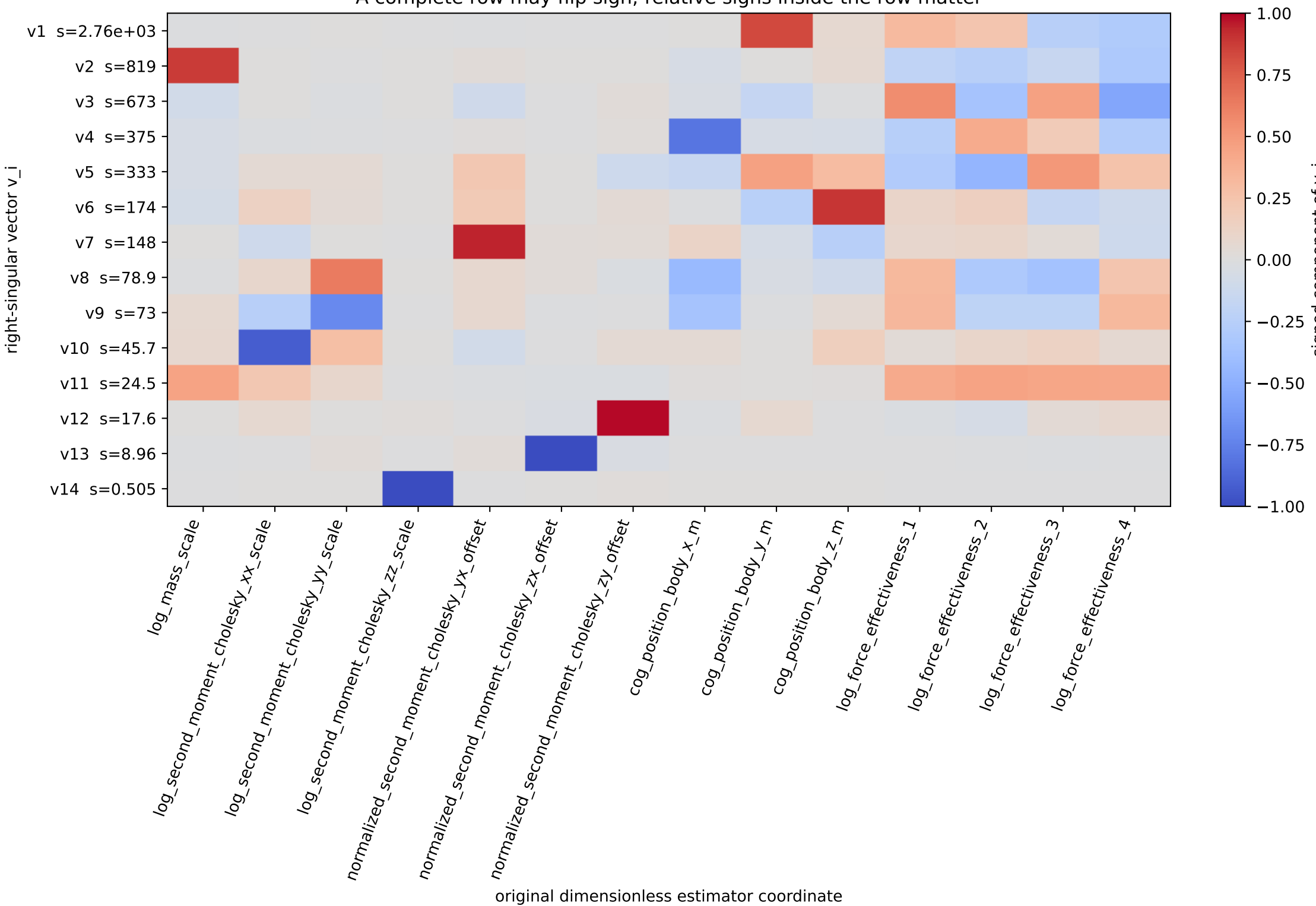
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.38050524069

inertia J [kg m²]

[0.064965321 -4.8945501e-05 -2.0424756e-05]
[-4.8945501e-05 0.065257535 0.00010650398]
[-2.0424756e-05 0.00010650398 0.12895152]

principal moments [kg m²]

[0.06495734, 0.065265331, 0.1289517]

CoG position in body frame [m]

[-0.002154224, 0.0011196859, 0.010937152]

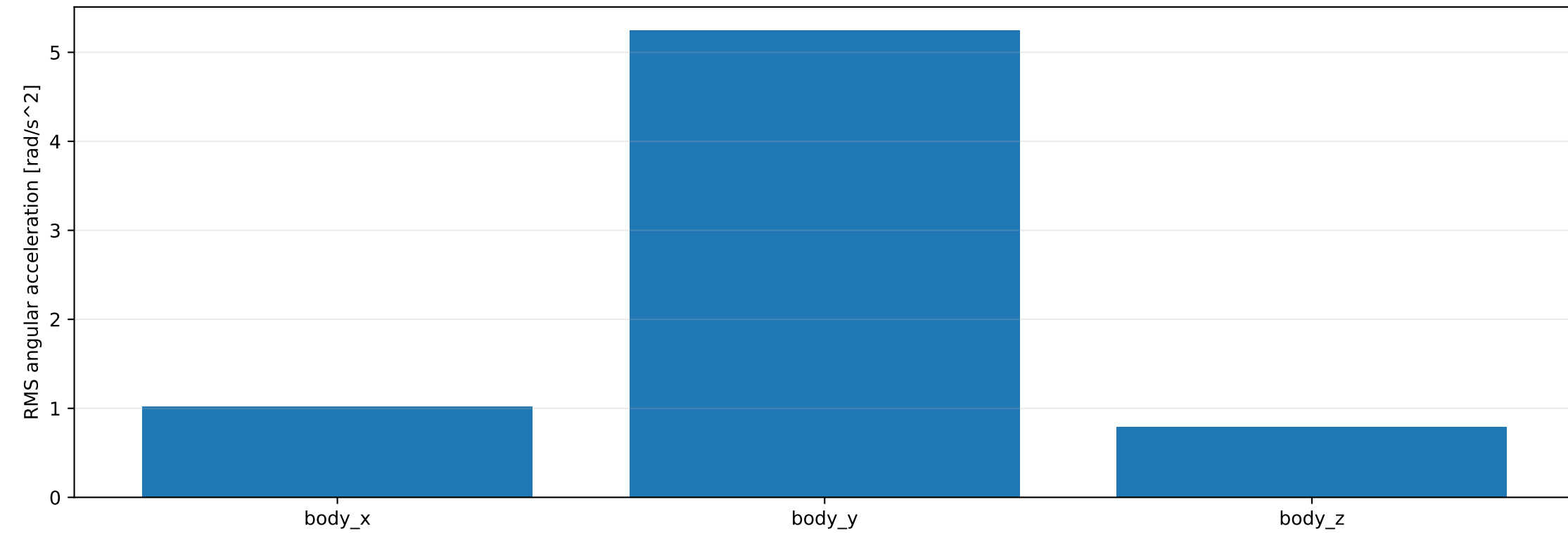
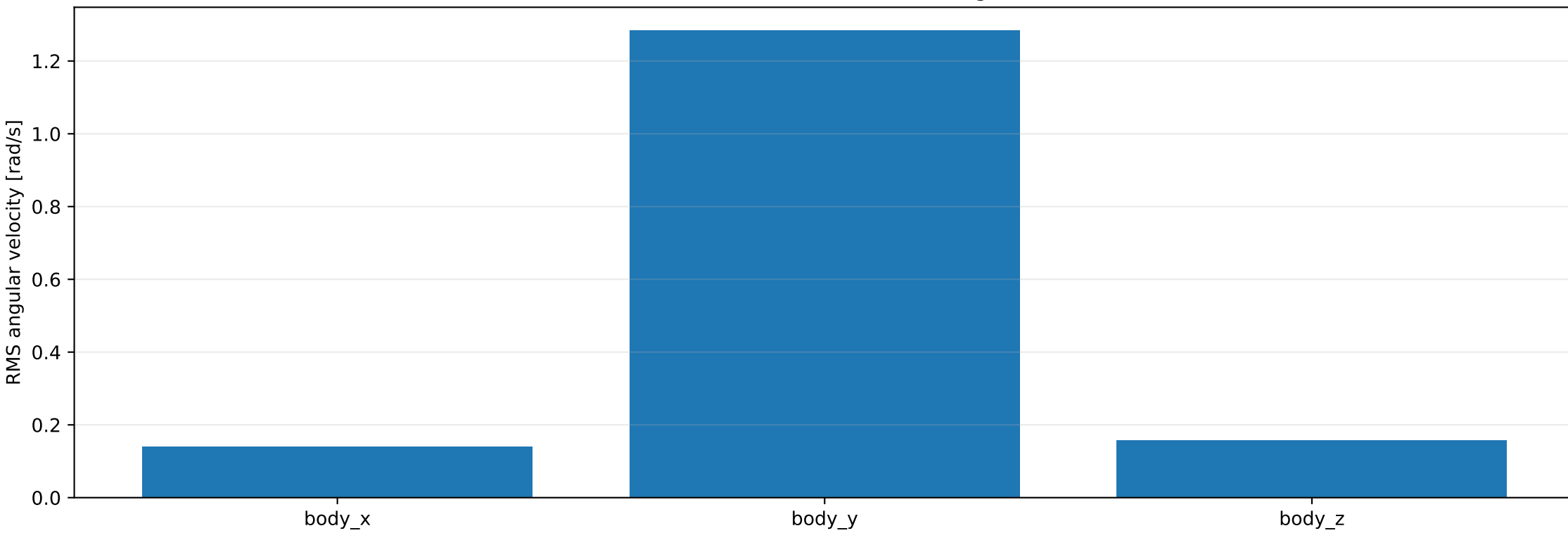
rotor force effectiveness

[0.89612591, 0.91216521, 0.88417985, 0.88653425]

selected delay [s]

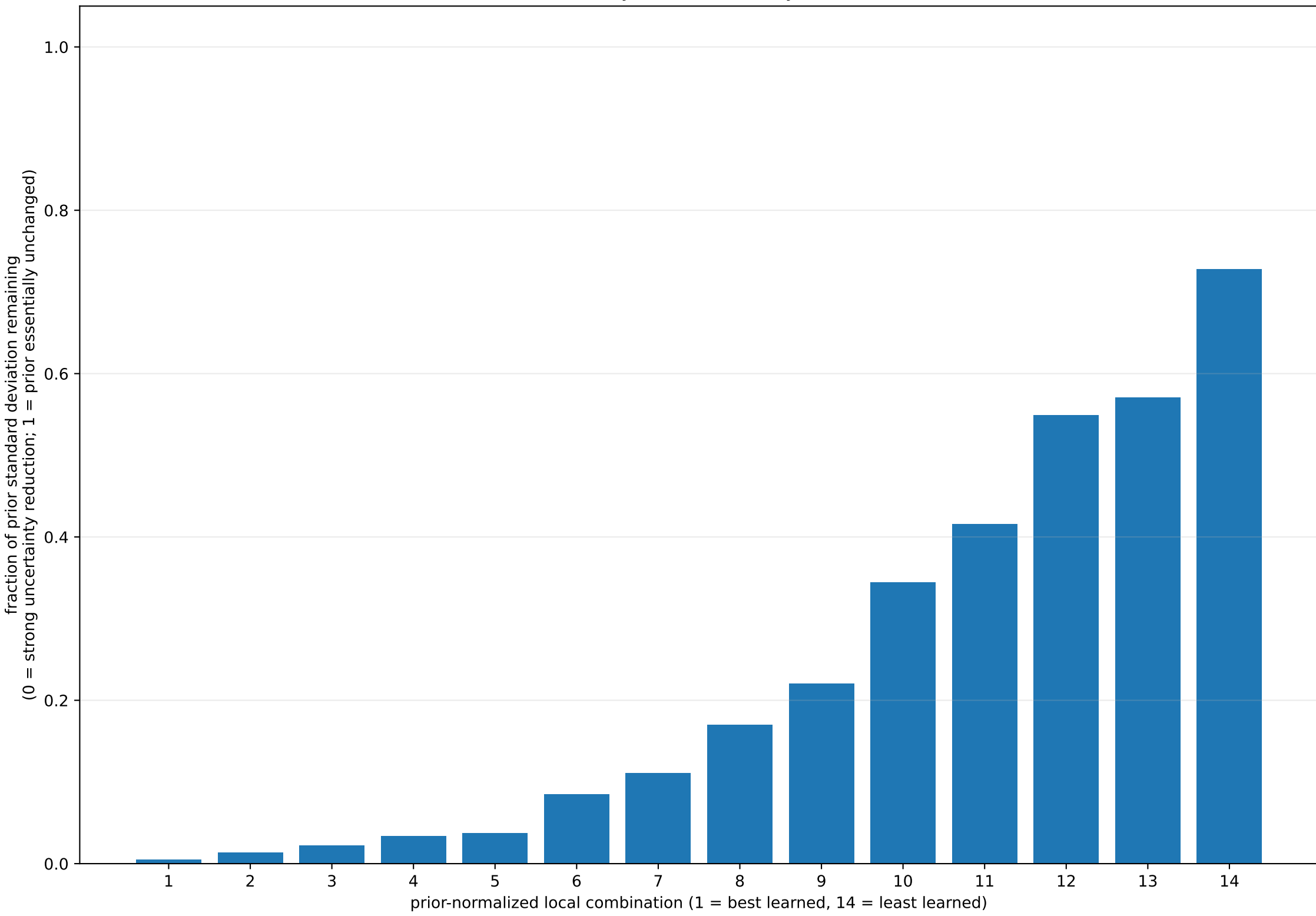
0.038662298837

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

